

3. Methodology and Evaluation Framework

3.1 - Objectives and Methodological Framework

The method works backward from the intended outcome: stakeholders should be able to examine attributed cloud costs through understandable indicators. The research question in section 1.3 is translated into capabilities, artifacts and acceptance criteria. The table follows this deduction down to its prerequisite, a suitable dataset; it does not describe the execution schedule or claim that every criterion has been met.

Each identified need is translated into an objective, a proposed artifact and a criterion for validation:

Need → objective → proposed response/solution: an artifact → validation criterion

Intended outcome	Required capability	Project artifact	Acceptance criteria
Support FinOps decisions	Integrate analysis and operational monitoring	FinOps data platform	Indicator meanings and data provenance are identifiable
Provide stakeholder showback	Publish understandable, authorized cost views	Dashboard and documented KPIs	Business questions can be answered; totals and access boundaries are checked
Ensure consistent calculations	Define grain, measures and aggregation rules	Analytical model, dictionary and datamarts	Questions map to data; calculations preserve grain and reference totals
Maintain trust as sources evolve	Control quality, responsibilities and changes	Versioned Data Contract and quality reports	Incompatible inputs are detected; invalid publication is blocked and explained
Keep products updated	Automate processing and recovery	Daily/Monthly jobs and audit records	Expected periods are processed; replay avoids duplicates; failures are diagnosable
Establish feasibility	Test the complete chain locally	End-to-end POC	Source, transformed and displayed values agree
Supply experimental inputs	Generate reproducible normal and correction cases	Synthetic FOCUS corpus, generator and manifests	Schema, coverage, totals and scenario expectations are verified

Implementation proceeds from the corpus and local POC to automated cloud processing and online consumption, with quality, governance and security accompanying each stage. Repeated build-and-test cycles are consistent with the FinOps Foundation's FOCUS adoption

guidance (FinOps Foundation, n.d.-a). The sections below explain this method; architecture, implementation and evaluation are developed in chapters 4, 5, 7 and 8.

Source for this section (accessed 30 September 2026):

- FinOps Foundation, *Adopting FOCUS, the FinOps Open Cost and Usage Specification*: <https://www.finops.org/wg/adopting-focus-the-finops-open-cost-and-usage-specification/>

3.2 - Experimental Dataset Preparation

3.2.1 Reference data and source preparation

Two Azure FOCUS cost-and-usage files, anonymized before use, provide reference rows and schema for fictional consumption data. Microsoft's FOCUS file-schema documentation defines the fields but does not establish the provenance of these particular files (Microsoft, n.d.-a). The reference inputs remain local and are not published with the project.

The independent generator retains rows with usable charge dates and billed cost, EUR currency and Microsoft provider. The `daily-template-v2` method samples these rows, replaces selected identifiers consistently, assigns dates and scales costs to configured targets. Daily output retains the source Arrow schema and is written to `datasets/focus/daily/YYYY/MM/YYYY-MM-DD.parquet`. The same published inputs can subsequently feed different analytical engines.

A date-specific seed makes sampling independent of batch boundaries. The configuration sets an annual `BilledCost` target of EUR 12 million for 2025, monthly weights and a 1.06 growth multiplier for 2026. Targets are distributed across days in cents. These are modeling assumptions, not estimates of observed seasonality or company spending.

A configuration-and-source fingerprint and daily ledger record paths, rows, amounts and SHA-256 hashes. Resumption checks existing entries and files rather than silently replacing them. Stable synthetic identifiers preserve selected links; their substitution alone does not prove that every potentially identifying attribute has been removed.

3.2.2 Monthly billing and controlled scenarios

A monthly close needs a billing reference distinct from provisional Daily usage. Without a real invoice feed, the simulator constructs this reference from a complete month of Daily files. It requires every calendar day, matching ledger entries and hashes, consistent schemas and periods, and agreement on rows, currency, provider and `BilledCost`.

The `no_change` scenario concatenates rows unchanged. Other scenarios inject known differences: `late_usage` adds a EUR 1,500 Usage line; `cost_correction` reduces an eligible line by EUR 200; `combined` applies both; and `unexplained_difference` changes a line by EUR 50 without an explanation. Deterministic selection makes these interventions repeatable.

The manifest records affected rows, before/after amounts and expected differences. It serves as an evaluation oracle for detected changes, missed changes and blocking decisions, but must not inform the pipeline's decision to close a month. Financial acceptance rules require FinOps review; a known injected amount is not automatically an acceptable difference.

Monthly output is published locally at `datasets/focus/monthly/billing-YYYY-MM.parquet`, with a separate manifest. Read-back checks precede publication; explicit replacement archives the previous version. Daily and Monthly represent the same month's activity, so adding their amounts would double-count it.

3.2.3 Coverage and limitations

Local validation on 27 September 2026 confirmed schema, temporal and monetary consistency for 608 Daily files, January 2025-August 2026, containing 3,664,261 rows and EUR 20,357,040. The 18 `no_change` Monthly files cover January 2025-June 2026, with 3,279,613 rows and EUR 18,220,080; each matched its Daily month and manifest hash. Revalidation after path harmonization on 28 September preserved these results and the Monthly hashes.

July-August therefore belong only to the validated Daily corpus. Corrected scenarios still require downstream tests against expected outcomes. The corpus supports engineering evaluation, not claims about real resource lifecycles, invoice correctness or organizational savings; forecasting would assess behavior under the generator's assumptions.

Source for this section (accessed 30 September 2026):

- Microsoft, *FOCUS cost and usage details file schema*: <https://learn.microsoft.com/en-us/azure/cost-management-billing/dataset-schema/cost-usage-details-focus>

3.3 - Data Quality, Governance and Change Control

Quality controls cover schema, required values, periods, monetary reconciliation and repeatability. Technical validity is assessed separately from business attribution: a reconciled total can still lack an application or owner. Missing allocation attributes remain visible, and negative amounts are not rejected solely by sign because they may represent credits. EUR and Microsoft delimit this corpus, not all FOCUS inputs.

The versioned Data Contract declares fields, types, nullability and selected rules. Failed validation must block invalid canonical publication and identify the input and rule for diagnosis. Passing this project contract is not certification of complete FOCUS conformance. Governance also requires traceable revisions, corrections and access decisions. The proposed responsibilities assign business definitions to FinOps and technical controls to data engineering; organizational confirmation remains necessary.

Specification and contract versions are explicit because FOCUS version numbers do not guarantee compatibility, as stated in the official repository (FinOps Open Cost and Usage Specification, n.d.). The project's compatibility policy permits additional fields, but rejects missing required fields and incompatible type changes. Intentional evolution requires review of field meanings, a versioned adaptation and regression tests of dependent calculations. Schema detection does not automatically revise the contract or detect every semantic change.

Access tests cover files, queries, dashboards and exports. Public synthetic demonstrations and authorized stakeholder services are assessed separately; chapter 7 reports the available security evidence.

Source for this section (accessed 30 September 2026):

- FinOps Open Cost and Usage Specification, *FOCUS_Spec repository*, “Versioning the Specification”: https://github.com/FinOps-Open-Cost-and-Usage-Spec/FOCUS_Spec#versioning-the-specification

3.4 - Progressive Implementation and Automation

The local POC tests agreement between source totals, transformations and displayed indicators for an identified period. The cloud extension then addresses recurring execution, separate development and production outputs, and remote consumption. This explains the implementation strategy without assuming a strict chronology or treating local success as cloud validation.

The operating protocol distinguishes historical backfill, incremental Daily processing and monthly replacement. Each workflow defines input scope, expected outputs and replay behavior. Development validation precedes production promotion; source identity, status and reconciliation results must support recovery if one environment succeeds and the next fails. Repeated execution must not duplicate activity.

Evaluation follows the acceptance criteria in section 3.1. Each test records its inputs, period, environment, revision, procedure, expected result and observed output, including failures. Cloud records also identify runs and relevant table states. Code demonstrates implemented logic; retained outputs establish behavior in a particular environment. Chapter 7 separates available results from unresolved tests.

Performance and cost comparisons require equivalent workloads and verified outputs, with compute, cache and measurement boundaries recorded. Shared and idle intervals need explicit attribution. Otherwise the comparison is an operational observation rather than a controlled benchmark; chapter 8 develops this protocol.

3.5 - Analytical Preparation

Each business question determines a measure, population, period and allocation scope. For example, monthly cost by application requires a stated cost basis and attribution rule. Indicator specifications record source fields, formula, unit, grain, filters, exclusions and missing-value treatment, using the financial definitions in chapter 2.

These specifications guide the dimensional model and subject-oriented datamarts. Joins and aggregations must preserve the intended charge population without multiplying costs. Each datamart is compared with a reference calculation on the same scope; displayed indicators are then checked against the datamart read by the dashboard. Section 4.5 defines the concrete model and maps its products to business questions, while chapter 6 examines their analytical interpretation.

3.6 - FinOps Data Product Delivery and Operation

The dashboard is treated as a stakeholder-facing analytical data product with defined questions, consumers, measures and maintenance expectations. This use of the term follows Google Cloud's definition, which includes dashboards (Google Cloud, n.d.). Datamarts are reusable analytical products; the platform provides their production, validation and distribution.

Delivery requires an identifiable deployed version, verified permissions and visibility of the last successful update, covered periods and known quality limitations. Issues must be traceable to inputs, rules and runs. The PFE's production environment is not equated with enterprise deployment.

Acceptance combines calculation and access checks with defined user tasks, such as identifying a cost driver or examining allocation. Results are compared with reference outputs and representative feedback where available. Technical success alone does not demonstrate stakeholder adoption, reduced effort or realized savings; those claims require separate evidence.

Source for this section (accessed 30 September 2026):

- Google Cloud, *What are data products?*:
<https://cloud.google.com/discover/what-are-data-products>

References

FinOps Foundation. (n.d.-a). *Adopting FOCUS, the FinOps Open Cost and Usage Specification*.

<https://www.finops.org/wg/adopting-focus-the-finops-open-cost-and-usage-specification/> (last updated 10 December 2025; accessed 30 September 2026).

FinOps Open Cost and Usage Specification. (n.d.). *FOCUS_Spec GitHub repository*, “Versioning the Specification”. https://github.com/FinOps-Open-Cost-and-Usage-Spec/FOCUS_Spec#versioning-the-specification (accessed 30 September 2026).

Google Cloud. (n.d.). *What are data products?* <https://cloud.google.com/discover/what-are-data-products> (accessed 30 September 2026).

Microsoft. (n.d.-a). *FOCUS cost and usage details file schema*.

<https://learn.microsoft.com/en-us/azure/cost-management-billing/dataset-schema/cost-usage-details-focus> (accessed 30 September 2026).